(2) However, the differential equation $\frac{dy}{dx}=\frac{x^3+y^2}{2x^3-xy^2}$ is not homogeneous. (verify!)

To find the solution of a homogeneous differential equation $\frac{dy}{dx}=g\left(\frac{y}{x}\right)$, consider the substitution $v=\frac{y}{x}$. Then, $y=xv$ and $\frac{dy}{dx}=v+x\frac{dv}{dx}$. Thus, the given differential equation becomes $x\frac{dv}{dx}=f(v)-v$ which is solved using variables separable method. This leads to the following result.

Theorem 10.1

If $M(x,y)dx+N(x,y)dy=0$ is a homogeneous equation, then the change of variable $y=vx$, transforms into a separable equation in the variables v and x.

Example 10.17

Solve $\left(x^2-3y^2\right)dx+2xydy=0$.

Solution

We know that the given equation is homogeneous.

Now, we rewrite the given equation as $\frac{dy}{dx}=\frac{3y}{2x}-\frac{x}{2y}$.

Taking $y=vx$, we have $v+x\frac{dv}{dx}=\frac{3v}{2}-\frac{1}{2v}$ or $x\frac{dv}{dx}=\frac{v^2-1}{2v}$.

Separating the variables, we obtain $\frac{2vdv}{v^2-1}=\frac{dx}{x}$.

On integration, we get $\log\left|v^2-1\right|=\log|x|+\log|C|$,

Hence $\left|v^2-1\right|=|Cx|$, where C is an arbitrary constant.

Now, replace v by $\frac{y}{x}$ to get $\left|\frac{y^2}{x^2}-1\right|=|Cx|$.

Thus, we have $\left|y^2-x^2\right|=\left|Cx^3\right|$.

Hence, $y^2-x^2=\pm Cx^3$ (or) $y^2-x^2=kx^3$ gives the general solution.

Example 10.18

Solve $\left(y+\sqrt{x^2+y^2}\right)dx-xdy=0,\quad y(1)=0$.

Solution

The given differential equation is homogeneous (verify!).

Now, we rewrite the given equation in differential form $\frac{dy}{dx}=\frac{y+\sqrt{x^2+y^2}}{x}$.

Since the initial value of x is 1, we consider $x>0$ and take $x=\sqrt{x^2}$.

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